

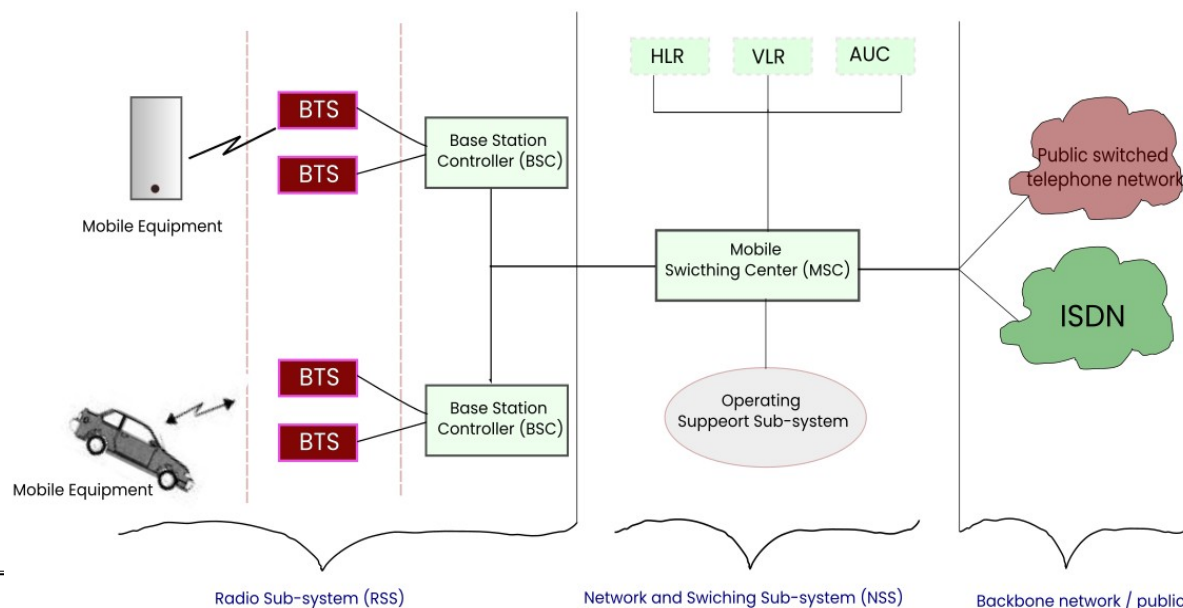
A Survey: Generation of Mobile Communication System

1. The Second Generation (2G)

Global system for mobile communication (GSM) is a globally accepted standard for digital cellular communication. GSM is the name of a standardization group established in 1982 to create a common European mobile telephone standard that would formulate specifications for a pan-European mobile cellular radio system operating at 900 MHz. It is estimated that many countries outside of Europe will join the GSM partnership.

GSM provides recommendations, not requirements. The GSM specifications define the functions and interface requirements in detail but do not address the hardware. The reason for this is to limit the designers as little as possible but still to make it possible for the operators to buy equipment from different suppliers.

The GSM network is divided into three major systems: The Network and switching subsystem system (NSS), the base station system (BSS), and the operation and support system (OSS).



- **Radio Subsystem (RSS):** comprises all radio specific entities, i.e., the mobile stations (MS) and the base station subsystem (BSS). A connection between the RSS and the NSS via the A interface (solid lines) and the connection to the OSS via the O interface.
- **Base station subsystem (BSS):** A GSM network comprises many BSSs, each controlled by a base station controller (BSC). The BSS performs all functions necessary to maintain radio connections to an MS, coding/decoding of voice, and rate adaptation to/from the wireless network part. Besides a BSC, the BSS contains several BTSs.
- **Base station controllers (BSC):** The BSC provides all the control functions and physical links between the MSC and BTS. It is a high capacity switch that provides functions such as handover, cell configuration data, and control of radio frequency (RF) power levels in BTS. A number of BSC's are served by and MSC.
- **Base transceiver station (BTS):** The BTS handles the radio interface to the mobile station. A BTS can form a radio cell or, using sectorized antennas, several others and is connected to MS via the U_m interface, and to the BSC via the Abis interface. The Um interface contains all the mechanisms necessary for wireless transmission (TDMA, FDMA etc.) The BTS is the radio equipment (transceivers and antennas) needed to service each cell in the network. A group of BTS's are controlled by an BSC.
- **Network Switching Subsystem:** The NSS is responsible for performing call processing and subscriber related functions. The switching system includes the following functional units: One of the heart component is of this sub-system is Mobile Switching center. Which performs the telephony switching functions of the system. It controls calls to and from other telephone and data systems.
- **Home location register (HLR):** It is a database used for storage and management of subscriptions. HLR stores permanent data about

subscribers, including a subscribers service profile, location information and activity status. When an individual buys a subscription from the PCS provider, he or she is registered in the HLR of that operator.

- Visitor location register (VLR): It is a database that contains temporary information about subscribers that is needed by the MSC in order to service visiting subscribers. VLR is always integrated with the MSC. When a MS roams into a new MSC area, the VLR connected to that MSC will request

data about the mobile station from the HLR.

- Authentication center (AUC): A unit called the AUC provides authentication and encryption parameters that verify the users identity and ensure the confidentiality of each call.
- Equipment identity register (EIR): It is a database that contains information about the identity of mobile equipment that prevents calls from stolen, unauthorized or defective mobile stations.

➤ Operation and Support system: The operations and maintenance center (OMC) is connected to all equipment in the switching system and to the BSC. Implementation of OMC is called operation and support system (OSS). The OSS is the functional entity from which the network operator monitors and controls the system. The purpose of OSS is to offer the customer cost-effective support for centralized, regional and local operational and maintenance activities that are required for a GSM network. OSS provides a network overview and allows engineers to monitor, diagnose and troubleshoot every aspect of the GSM network.

The mobile station (MS) consists of the [mobile equipment](#) and a smart card called the [Subscriber Identity Module \(SIM\)](#). The SIM provides personal mobility, so that the user can have access to subscribed services irrespective of a specific terminal. By inserting the SIM card into another GSM terminal, the user is able to receive calls at that terminal, make calls from that terminal, and receive other subscribed services.

The mobile equipment is uniquely identified by the International Mobile Equipment Identity (IMEI). The [SIM card contains the International Mobile Subscriber Identity \(IMSI\) used to identify the subscriber to the system](#), a secret key for authentication, and other information. The IMEI and the IMSI are independent, thereby allowing personal mobility. The SIM card may be protected against unauthorized use by a password or personal identity number.

GSM Protocols

The signalling protocol in GSM is structured into three general layers depending on the interface.

[The first Layer is the physical layer](#) that handles all radio-specific functions. This includes the creation of bursts according to the five different formats, [multiplexing of bursts into a TDMA frame](#), [synchronization with the BTS](#), [detection of idle channels](#), and [measurement of the channel quality on the downlink](#).

The physical layer at U_m uses [GMSK](#) for digital modulation and performs encryption/decryption of data, i.e., encryption is not performed end-to-end, but only between MS and BSS over the air interface. Protocol architecture for Signaling

The main tasks of the physical layer comprise channel coding and error detection/correction, which is directly combined with the coding mechanisms. Channel coding makes extensive use of different forward error correction (FEC) schemes.

Signaling between entities in a GSM network requires higher layers. For this purpose, the LAPD_m protocol has been defined at the U_m interface for layer two. LAPD_m has been derived from link access procedure for the D-channel (LAPD) in ISDN systems, which is a version of HDLC. LAPD_m is a lightweight LAPD because it does not need synchronization flags or checksumming for error detection. LAPD_m offers reliable data transfer over connections, re-sequencing of data frames, and flow control.

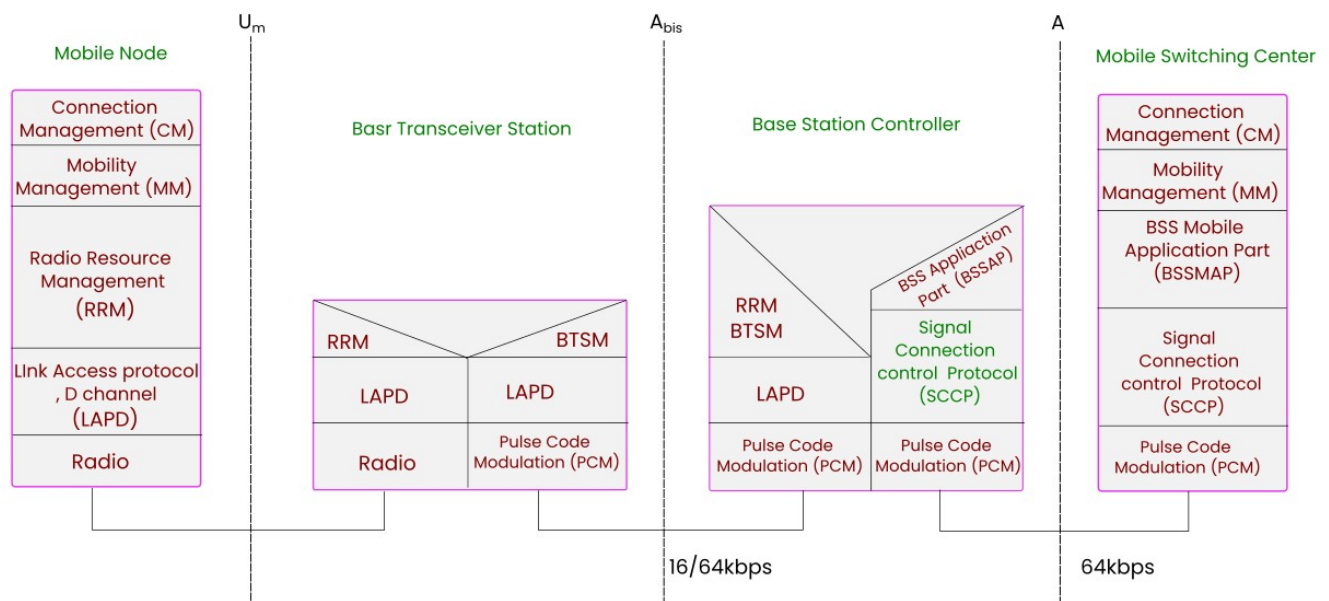
The network layer in GSM, layer three, comprises several sub-layers. The lowest sub-layer is the radio resource management (RR). Only a part of this layer, RR', is implemented in the BTS, the

remainder is situated in the BSC. The functions of RR' are supported by the BSC via the BTS management (BTSM). The main tasks of RR are setup, maintenance, and release of radio channels. Mobility management (MM) contains functions for registration, authentication, identification, location updating, and the provision of a temporary mobile subscriber identity (TMSI).

Finally, the call management (CM) layer contains three entities: call control (CC), short message service (SMS), and supplementary service (SS). SMS allows for message transfer using the control channels SDCCH and SACCH, while SS offers the services like user identification, call redirection, or forwarding of ongoing calls. CC provides a point-to-point connection between two terminals and is used by higher layers for call establishment, call clearing and change of call parameters. This layer also provides functions to send in-band tones, called

dual tone multiple frequency (DTMF), over the GSM network. These tones are used, e.g., for the remote control of answering

machines or the entry of PINs in electronic banking and are, also used for dialing in traditional analog telephone systems.



There are two basic types of services offered through GSM:

1. Telephony (also referred to as teleservices)

Telephony services are mainly voice services that provide subscribers with the complete capability (including necessary terminal equipment) to communicate with other subscribers. Data services provide the capacity necessary to transmit appropriate data signals between two access points creating an interface to the network. In addition

to normal telephony and emergency calling, the following subscriber services are supported by GSM:

- [Dual-tone multifrequency \(DTMF\)](#)—DTMF is a tone signaling scheme often used for various control purposes via the telephone network, such as remote control of an answering machine. GSM supports full-originating DTMF.
- [Short message services](#)—A convenient facility of the GSM network is the short message service. A message consisting of a maximum of 160 alphanumeric characters can be sent to or from a mobile station. This service can be viewed as an advanced form of alphanumeric paging with a number of advantages. If the subscriber's mobile unit is powered off or has left the coverage area, the message is stored and offered back to the subscriber when the mobile is powered on or has reentered the coverage area of the network. This function ensures that the message will be received.
- [Cell broadcast](#)—A variation of the short message service is the cell broadcast facility. A message of a maximum of 93 characters can be broadcast to all mobile subscribers in a certain geographic area. Typical applications include traffic congestion warnings and reports on accidents.
- [Voice mail](#)—This service is actually an answering machine within the network, which is controlled by the subscriber. Calls can be forwarded to the subscriber's voice-mail box and the subscriber checks for messages via a personal security code.
- [Fax mail](#)—With this service, the subscriber can receive fax messages at any fax machine. The messages are stored in a service center from which they can be retrieved by the subscriber via a personal security code to the desired fax number.

2. Data Services / bearer services

GSM mainly focuses on **voice-oriented tele services**. These comprise encrypted voice transmission, message services, and basic data communication with terminals as known from the PSTN or ISDN. The primary goal of GSM was the provision of high-quality digital voice transmission. Special codecs (coder/decoder) are used for voice transmission, while other codecs are used for the transmission of analog data for communication with traditional computer modems used in, e.g., fax machines. Another service offered by GSM is the emergency number (eg 911, 999, 994). This service is mandatory for all providers and free of charge. This connection also has the highest priority, possibly pre-empting other connections, and will automatically be set up with the closest emergency center. A useful service for very simple message transfer is the short message service (SMS), which offers transmission of messages of up to 160 characters. Sending and receiving of SMS is possible during data or voice transmission. It can be used for "serious" applications such as displaying road conditions, e-mail headers or stock quotes, but it can also transfer logos, ring tones, horoscopes and love letter

3. Supplementary Services

GSM supports a comprehensive set of supplementary services that can complement and support both telephony and data services. Supplementary services are defined by GSM and are characterized as revenue-generating features. A partial listing of supplementary services follows.

Call forwarding: → This service gives the subscriber the ability to forward

incoming calls to another number if the called mobile unit is not reachable, if it

is busy, if there is no reply, or if call forwarding is allowed unconditionally.

Barring of outgoing call → This service makes it possible for a mobile subscriber to prevent all outgoing calls.

Barring of incoming call→ This function allows the subscriber to prevent incoming calls. The following two conditions for incoming call barring exist: barring of all incoming calls and barring of incoming calls when roaming outside the home Net.

Advice of charge (AoC)—The AoC service provides the mobile subscriber with an estimate of the call charges. There are two types of AoC information: one that provides the subscriber with an estimate of the bill and one that can be used for immediate charging purposes. AoC for data calls is provided on the basis of time measurements.

Call hold → This service enables the subscriber to interrupt an ongoing call and then subsequently reestablish the call. The call hold service is only applicable to normal telephony.

Call waiting→ This service enables the mobile subscriber to be notified of an incoming call during a conversation. The subscriber can answer, reject, or ignore the incoming call. Call waiting is applicable to all GSM telecommunications services using a circuit-switched connection.

Multiparty service → The multiparty service enables a mobile subscriber to establish a multiparty conversation—that is, a simultaneous conversation between three and six subscribers. This service is only applicable to normal telephony.

Calling line identification presentation/restriction → These services supply the called party with the integrated services digital network (ISDN) number of the calling party. The restriction service enables the calling party to restrict the presentation. The restriction overrides the presentation.

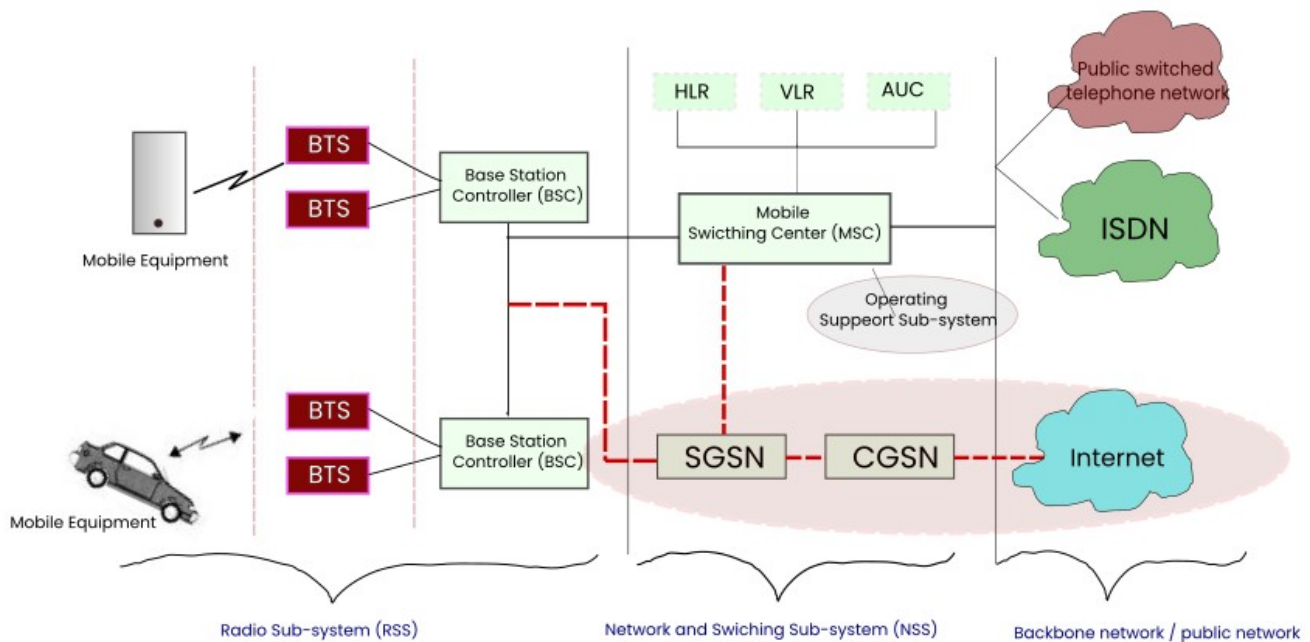
Closed user groups (CUGs)—CUGs are generally comparable to a PBX. They

are a group of subscribers who are capable of only calling themselves and certain numbers.

2. The Second and half Generation (2.5G)

2.5 G wireless cellular networks have been developed as a transition path to 3G. The notable are GPRS and EDGE. These systems build packet-switching systems on top of existing 2G systems to improve data rates significantly.

- GPRS (General Packet Radio Service): GSM networks offer circuit-switched data services at 9.6 Kbps. Most GSM carriers are developing a service called General Packet Radio Service (GPRS), a 2.5G technology. GPRS can theoretically provide IP-based packet data speeds up to a maximum of 28 Kbps and higher. However, typical GPRS networks operate at lower data rates. One proposed configuration is 80 Kbps maximum (56 Kbps typical) for the downlink and 20 Kbps maximum (14.4 Kbps typical) for the uplink. GPRS supports both IP and X.25 networking. GPRS can be added to GSM infrastructures quite readily. It takes advantage of existing 200 kHz radio channels and does not require new radio spectrum. GPRS basically overlays a packet switching network on the existing circuit switched GSM network. This gives the user an option to use a packet-based data service. An architectural view of GPRS Network Is Depicted as follows.



The two main component or addition of GPRS is [Gateway GPRS Support Node \(GGSN\)](#) and [Serving GPRS Support Node \(SGSN\)](#)

[Gateway GPRS Support Node \(GGSN\)](#)

The Gateway GPRS Support Node acts as an interface and a router to external networks. It contains routing information for GPRS mobiles, which is used to tunnel packets through the IP based internal backbone to the correct Serving GPRS Support Node. The GGSN also collects charging information

connected to the use of the external data networks and can act as a packet filter for incoming traffic.

[Serving GPRS Support Node \(SGSN\)](#)

The Serving GPRS Support Node is responsible for authentication of GPRS mobiles, registration of mobiles in the network, mobility management, and collecting information on charging for the use of the air interface.

3. The Second and half Generations (2.75G)

The phase after GPRS is called Enhanced Data Rates for GSM Evolution (EDGE).

EDGE, generally considered a 3G technology, introduces new methods at the physical layer, including a new form of modulation (8 PSK) and different ways of encoding data to protect against errors. But the higher layer protocols stay the same. Thus EDGE can deliver maximum data rates up to 500Kbps using the same GPRS infrastructure (practical throughput may be only half the maximum rate).

EDGE has been designed to address some of the limitations of GPRS. GPRS impacts a network's existing cell capacity because voice and GPRS calls both use the same network resources. The extent of the impact

depends upon the number of time slots, if any, that are reserved for exclusive use of GPRS. In addition, GPRS actual data rates are much lower than advertised. Specifically, achieving the theoretical maximum GPRS data transmission speed of 172.2 Kbps would require a single user taking over all eight time slots without any error protection. Finally, GPRS is based on a modulation technique known as Gaussian minimum-shift keying (GMSK). EDGE is based on the eight-phase-shift keying (8 PSK) modulation that allows a much higher bit rate across the air interface. Since 8 PSK is also used in 3G, network operators need to incorporate it at some stage to make the transition to third generation mobile phone systems.

4. 3G - Third Generation

This generation set the standards for most of the wireless technology we have come to know and love. Web browsing, email, video downloading, picture sharing and other

Smartphone technology were introduced in the third generation. Introduced commercially in 2001, the goals set out for third generation mobile communication were

to facilitate greater voice and data capacity, support a wider range of applications, and increase data transmission at a lower cost.

The 3G standard utilizes a new technology called **UMTS** as its core network architecture – Universal Mobile Telecommunications System. This network combines aspects of the 2G network with some new technology and protocols to deliver a significantly faster data rate. Based on a set of standards used for mobile devices and mobile telecommunications use services and networks that comply with the International Mobile Telecommunications-2000 (**IMT-2000**) specifications by the International Telecommunication Union. One of requirements set by IMT-2000 was that speed should be at least 200Kbps to call it as 3G service.

3G has Multimedia services support along with **streaming** are more popular. In 3G, Universal access and portability across different device types are made possible (Telephones, PDA's, etc.). 3G increased the efficiency of frequency spectrum by

improving how audio is **compressed** during a call, so more simultaneous calls can happen in the same frequency range. The UN's International Telecommunications Union **IMT-2000** standard requires stationary speeds of 2Mbps and mobile speeds of 384kbps for a "true" 3G. The theoretical max speed for **HSPA+** is 21.6 Mbps

3G networks are wide area cellular telephone networks which evolved to incorporate high-speed internet access and video telephony. IMT-2000 defines a set of technical requirements for the realization of such targets, which can be summarized as follows:

- ◆ high data rates: 144 kbps in all environments and 2 Mbps in low-mobility and indoor environments.
- ◆ symmetrical and asymmetrical data transmission.
- ◆ Circuit-switched and packet-switched-based services
- ◆ Speech quality comparable to wire-line quality

- ◆ improved spectral efficiency
- ◆ several simultaneous services to end users for multimedia services
- ◆ seamless incorporation of second-generation cellular systems
- ◆ global roaming
- ◆ Open architecture for the rapid introduction of new services and technology.

3G Standards and Access Technologies

There are several different radio access technologies defined within ITU, based on either CDMA or TDMA technology. An organization called [3rd Generation Partnership Project](#) (3GPP) has continued that work by defining a mobile system that fulfills the IMT-2000 standard. This system is called Universal Mobile Telecommunications System (UMTS). After trying to establish a single 3G standard, ITU finally approved a family of five 3G standards, which are part of the 3G framework known as IMT-2000:

- ◆ W-CDMA
- ◆ CDMA2000
- ◆ TD-SCDMA

Europe, Japan, and Asia have agreed upon a 3G standard called the Universal Mobile Telecommunications System (UMTS), which is WCDMA operating at 2.1GHz. UMTS and WCDMA are often used as synonyms. In the USA and other parts of America, WCDMA will have to use another part of the radio spectrum

4. Evolution of 4G

The development of 4G Networks and related technologies in today's scenario is imperative indicator of advancement in the field of wireless

communication and technology. 4G will utilize most of the existing wireless communication infrastructure. The 4G Mobile communications will be based on the Open Wireless Architecture (OWA) to ensure the single terminal can seamlessly and automatically connect to the local high speed wireless access systems when the users are in the offices, homes, airports or shopping centers where the wireless access networks (i.e. Wireless LAN, Broadband Wireless Access, Wireless Local Loop, Home RF, Wireless ATM, etc) are available. When the users move to the mobile zone (i.e. Highway, Beach, Remote area, etc.), the same terminal can automatically switch to the wireless mobile networks (i.e. GPRS, W-CDMA, cdma2000, TD-SCDMA,

etc.). Based on this OWA model, 4G mobile will deliver the best business cases to the wireless and mobile industries, i.e. cdma2000/WLAN/ GPRS 3-in-1 product, WCDMA/OFDM/ WLAN 3-in-1 product, etc. The converged wireless communications can provide the following advantages:

1. Greatly increase the spectrum efficiency
2. Mostly ensure the highest data-rate to the wireless terminal
3. Best share the network resources and channel utilization
4. Optimally manage the service quality and multimedia applications with the

appropriate combination of resources; it is possible for 4G Networks to create alternatives that exceed consumer and industry expectations. Therefore, 4G developers must consider the appropriate security measures; the promotion of high-speed data transmission across the network, this 4G is intended to replace the current 3G systems within few years. The ambitious goal of 4G is to allow everyone to access the Internet anytime and everywhere. The provided connection to Internet will allow users to access all type of services including text, databases, and multimedia. 4G, unlike 3G, is IP based, that is every user connected to the Internet will have an IP address. This feature makes it easier to integrate the infrastructure of all current networks and consequently it will be easier for users to access services and applications regardless of the environment. 4G will also provide higher bandwidth, data rate, lower authentication overhead, and will ensure that the service is constantly provided to the user without any disruption. Another key feature of 4G networks is high level of user friendly customization. That is, each user can choose the preferred level of quality of service, radio environment; etc. Accessing 4G networks will be possible virtually by using any wireless device such as PDAs, cell phones, and laptops. In various forms

and at various levels connection with the network applications can be accomplished efficiently and correctly

Cellular Generations



High usability and global roaming: The end user terminals should be compatible with any technology, at anytime, anywhere in the world. The basic idea is that the user should be able to take his mobile to any place, for example, from a place that uses CDMA to another place that employs GSM. Multimedia support: The user should be able to receive high data rate multimedia services. This demands higher bandwidth and higher data rate. Personalization: This means that any type of person should be able to access the service. The service providers should be able to provide customized services to different type of users.

4G Issues and Advantages:

- ◆ Access
- ◆ Handoff
- ◆ Location co-ordination
- ◆ Resource co-ordination to add new user

- ◆ Support for quality of Service.
- ◆ Wireless Securities & Authentication.
- ◆ Network failure & backup.
- ◆ Pricing and billing.

Advantages

- ◆ Support for interactive multimedia voice, streaming video, internet & other broadband services.
- ◆ IP based mobile system.
- ◆ High speed, high capacity & low cost per bit.
- ◆ Gobal access, service portability & scalable mobile services.
- ◆ Better scheduling and call admission control technique.
- ◆ Ad-hoc & multi-hop network.
- ◆ Better spectral efficiency.
- ◆ Seamless network of multiple protocols & air interfaces.

Refer more on LTE and LTE advanced 4G standard

MOBILE CLOUDING COMPUTING

Mobile devices such as smartphones and tablets PCs are becoming an integral part of today 's lifestyle, as they are convenient and effective communication modes. There are many mobile applications that enrich the lives of the mobile users with many services, which can be found and downloaded from Google, and those made for the iPhone; these applications can run on many types of devices through a wireless network and remote servers. The swift development of mobile computing (MC) has become a forceful pattern in IT technology's development and in commerce and related fields. Nevertheless, the devices used to contain several challenges in terms of resources such as storage, bandwidth, and battery life, and communication in terms of security and mobility. These problems potentially impede the development of the quality of this service.

Cloud Computing (CC) is becoming popular tool for the computing architecture. CC provides several benefits for users of this service such as the infrastructure (networks, servers and storage) , platforms (operating systems and middleware) and software such as the many applications offered by the providers of cloud computing such as Google, Salesforce and Amazon. Moreover, CC allows the users to peruse the resources when it is needed

Mobile cloud computing (MCC) is the latest trend of cloud computing and is growing rapidly as a well utilized technology. The main aim is to use the techniques of cloud computing for the processing of data and storage in mobile devices. MCC has many benefits for both the mobile device users and enterprises.

The name of Mobile cloud computing was derived from the cloud computing concept. It has attracted the focus of business people as a viable business model

that lowers the cost of running or developing mobile application, the use of mobile devices as a form of technology that offers many mobile services at very low-cost

Definition

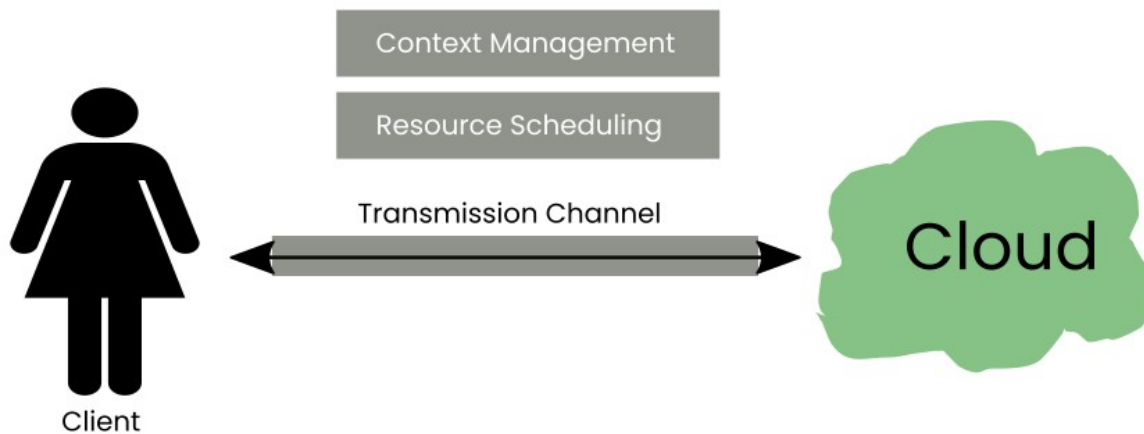
MCC infers that its a mixture of Cloud Computing and mobile web, which is common tool used by most mobile users when wanting to use Internet services and applications. In short, MCC offers cloud services such as storage services and data processing to mobile users. The mobile device used for this purpose does not have to be of high capacity in terms of memory and CPU speed since all the difficult modules of computing are taken care of in the cloud.

This section reviews the conceptual model that is used to examine the mobile cloud computing technology, and offers a few models of architecture for managing the Mobile Cloud Computing systems.

Jason H Christensen [31] proposed three types of the archetype for building the architecture for mobile applications of next generation's standards, which include the integration of smart mobile devices, context enabling and REST-based cloud computing. These three parts of the model are aligned to the transmission model in the mobile cloud which is the "Client-Connection-Cloud".

The cloud and client are shown to the right and the left. In between the cloud and the client side is the component called the "Transmission Channel". On this component are found the "Context Management" and the "Resource Scheduling" parts, which are on the cloud and client sides. This model has a prerequisite namely that the client is context-aware and that the Cloud side must provide a flexible, service on demand

for the client. The three middle portions in the model will be explained in a top-down approach. The following picture shows the abstract conceptual model of MCC



Transmission channel: The channel consists of different wireless communication protocols which mobile client can connect to the cloud by using these protocols.

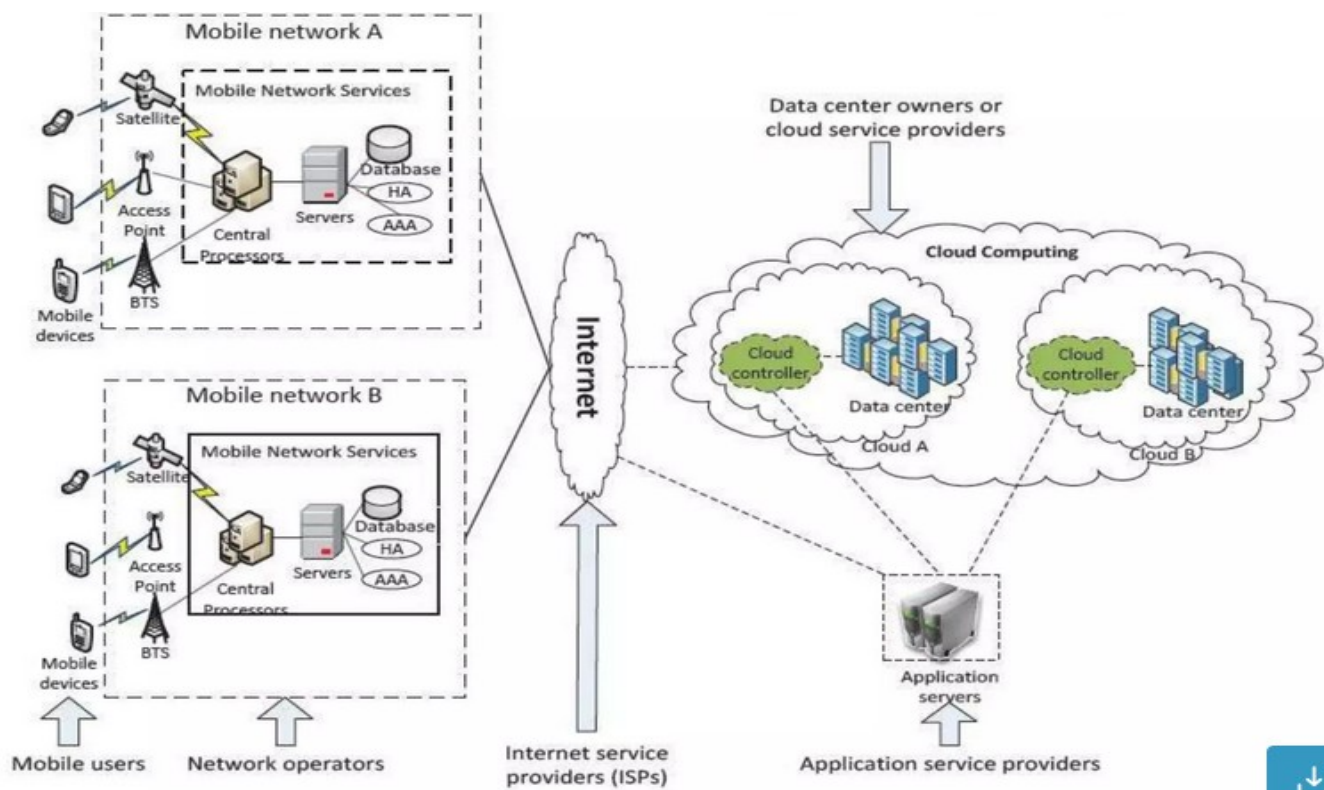
Resource scheduling: This component handles computer and storage resource by using virtual machines. The VM placement decisions are made by a separated scheduler component. It depends on data that is run on real time using VMs and the physical resources that are available.

Context Management: This module tracks context parameters and modify them according to context conditions. To have this kind of capability, some new applications such as explicit spatial contexts, spatial augmented reality (SAR), and Location Based Services (LBS) are highly used.

Architecture of Mobile Cloud Computing

This figure shows how the mobile devices are linked to the mobile networks through the use of the base stations, such as the base transceiver station, access point, or satellite. The information and requests (location and ID) made by the mobile users are channeled to the central processors which are linked to servers that provide the network services. From this place, the mobile network operators are able to offer services to users such as authorization, authentication, and accounting according to the home agent and data of the subscribers that is kept in the databases. Subsequently, the requests made by the subscribers are given to the cloud using the Internet service.

Cloud controllers inside the cloud will process the requests made and offer the requested services. The concepts of virtualization, service-oriented architecture, and utility computing (database servers, application, and web) are used to develop these services.



Advantage of Mobile Cloud Computing

- ◆ Extending battery lifetime.
- ◆ Improving data storage capacity and processing power
- ◆ Improving reliability
- ◆ Dynamic provisioning
- ◆ Scalability
- ◆ Flexible and portable
- ◆ On-demand services
- ◆ Cost effectiveness
- ◆ Universal access
- ◆ allows to make backup and recovery
- ◆ centralized monitoring and maintenance of software

****Review the disadvantage / challenge in Mobile cloud computing ****